Designer ODC Cargo Movement Report

Clean backend-rendered PDF for client sharing and underwriting review

Run Folder: voltas_as_on_20_03_2026__20260402_191659

Artifact Bundle: outputs__20260320_182304.zip

This report consolidates route concentration, vehicle concentration, carrier hotspots, consignee hotspots, and operational stress outputs from the ODC analytics notebook.

Total Exposure **Rs 118.62 Cr**	Trips **134**	Routes **41**
Carriers **13**	Vehicles **115**	Geo Coverage **92.54%**

Prepared with CargoShakti branding, watermarking, and print-safe layout. No browser print step is required.

CARGOSHA

Run tag: **20260402_191659**

Output folder: `outputs/voltas_as_on_20_03_2026__20260402_191659`

1. Executive Summary (1 page)

- **Total exposure analysed:** Rs 1.19B
- **Entities covered:** Routes=41 | Carriers=13 | Vehicles=115
- **Concentration insight (80/20 style):**
- Vehicle concentration: **Top 10 vehicles hold 27.28%** of total exposure (Top 1: 7.48%, Top 3: 14.82%).
- Route concentration: **Top 10 routes hold 68.67%** of total exposure.
- **Immediate underwriting red flags:**
- **Route risk hotspots**: highest risk-score routes include Vadodara → Cuttack, Vadodara → South West Delhi, Vadodara → Visakhapatnam.

2. Data Coverage & Quality

Quality gates (exported)

- CSV: `L0_quality_gates_summary__20260402_191659.csv`
- Summary table:

gate	status	notes	
:----------	:---------	:----------	
exposure_value_non_negative	PASS	negative rows: 0	
exposure_value_non_null	PASS	non-null: 134/134	
route_fields_present	PASS	present=['destination_city', 'origin_city']	
distance_km_non_null	PASS	non-null: 124/134	
distance_km_near_zero_check	PASS	near-zero (<5km) rows: 0 (intracity or same-city legs)	
duplicate_key_check	PASS	strict 12-digit eWayBill duplicated count: 0	missing/blank: 134
geo_columns_present	PASS	present=['origin_lat', 'origin_lon', 'dest_lat', 'dest_lon']	

Known blind spots

- Driver-level identifiers are not used as primary keys in this report (high churn, reassignment, and masking effects).

Why this data is still decision-grade

This dataset is decision-grade because it quantifies **where the money moves** (exposure concentration) and **where operational stress accumulates** (route/carrier/vehicle clusters). Even without claims, these are the right levers for underwriting controls: **sub-limits, deductibles, route conditions, carrier tiering, and inspection triggers**.

3. Route Risk Spine

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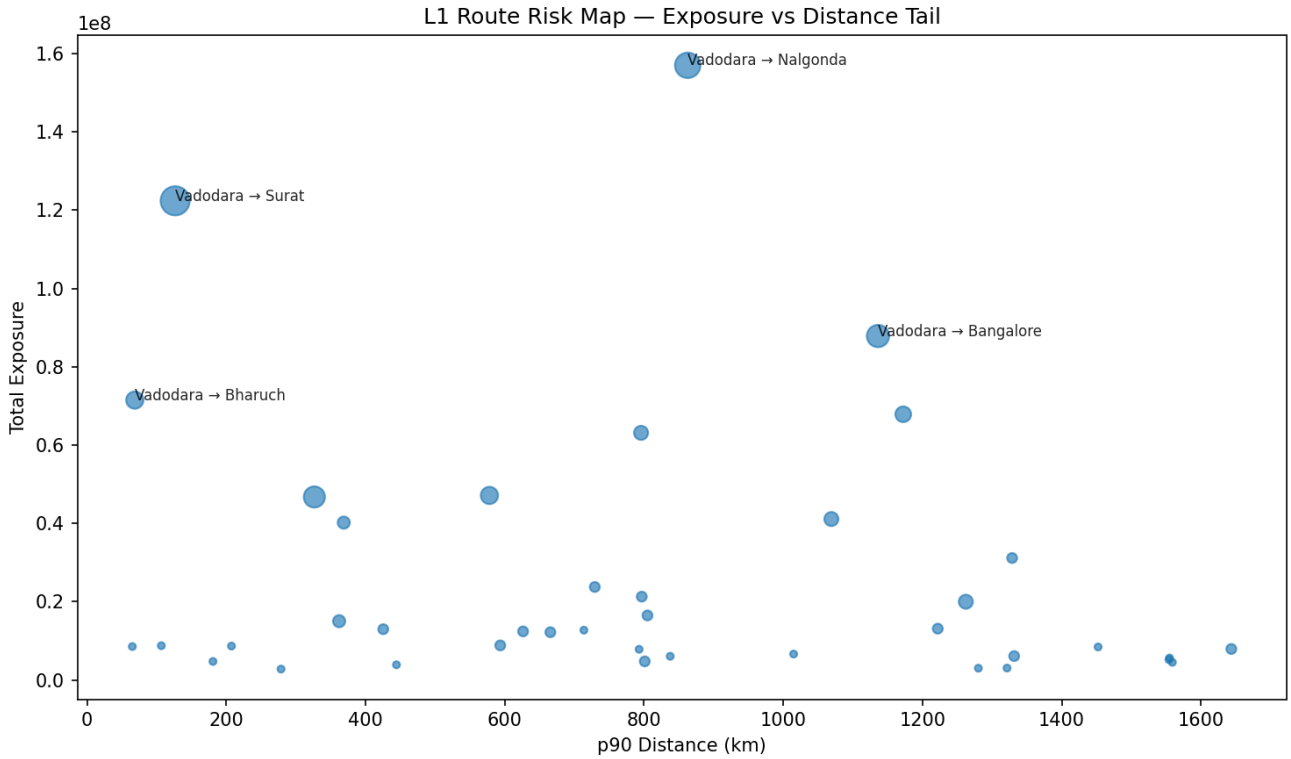
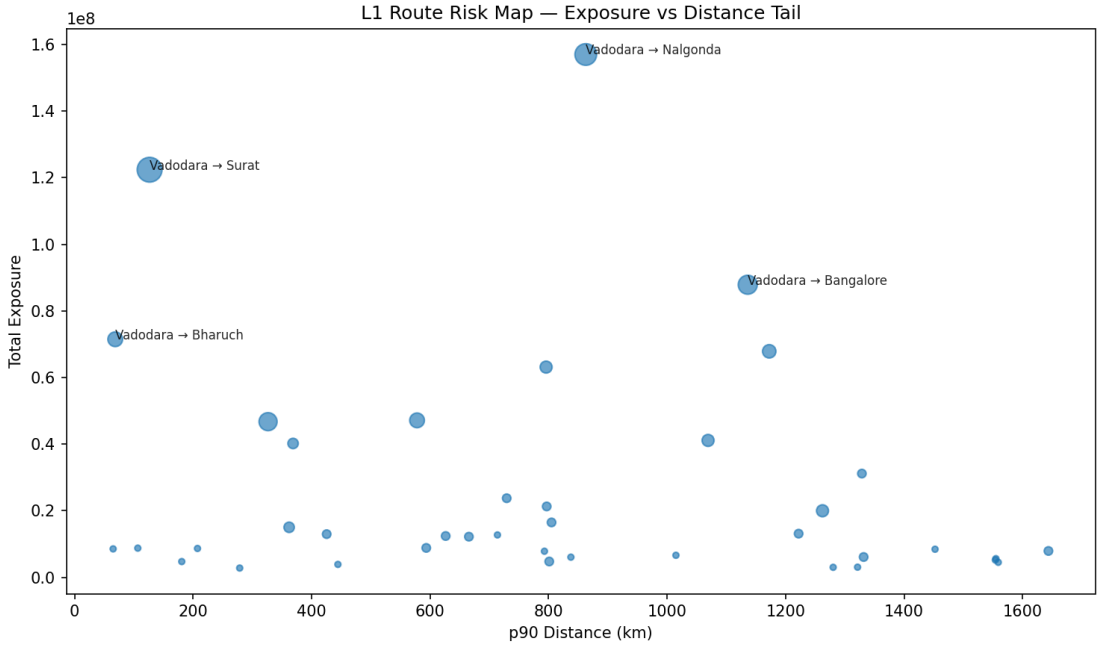


Route Risk Map (Auto-injected)

Top 10 risky routes (by risk score)



Route Risk Map (Auto-injected)

route	trip_count	total_exposure	avg_distance_km	p90_distance_km	exposure_per_trip	risk_score_0_100
:----------:	----------:	----------:	----------:	----------:	----------:	----------:
Vadodara → Cuttack	2	3.1152e+07	1636.22	1636.22	1.5576e+07	100
Vadodara → South West Delhi	4	6.31215e+07	1020.76	1020.76	1.57804e+07	94.0149
Vadodara → Visakhapatnam	5	6.78568e+07	1684.27	1684.27	1.35714e+07	85.4163

route	trip_count	total_exposure	avg_distance_km	p90_distance_km	exposure_per_trip	risk_score_0_100
Vadodara → Medak	1	1.27269e+07	953.903	953.903	1.27269e+07	71.781
Vadodara → Nalgonda	13	1.56975e+08	1166.92	1166.92	1.2075e+07	69.7198
Vadodara → Raigarh (MH)	3	4.01926e+07	498.492	498.492	1.33975e+07	67.661
Vadodara → Belgaum	2	2.37534e+07	959.447	959.447	1.18767e+07	65.96
Vadodara → East Godavari	4	4.1106e+07	1460.56	1460.56	1.02765e+07	59.2205
Vadodara → Dharwad	2	2.12872e+07	1050.59	1050.59	1.06436e+07	58.3838
Vadodara → Bangalore	10	8.78474e+07	1447.98	1447.98	8.78474e+06	48.1672

What underwriters should do

- Apply **route-tier rules**: higher deductible / stricter packing & escort rules on top-risk routes.
- Use **route pre-approval** for extreme-distance or extreme-exposure legs.
- Push **route survey / verification** for repeated high-exposure corridors.

4. Vehicle Concentration Risk

Why single-vehicle dominance is dangerous

When one vehicle (or a small set) carries disproportionate exposure, you create a **single-point-of-failure**. The operational system becomes brittle: breakdowns, detentions, driver churn, or route deviation spikes the insurer's tail risk.

Evidence from data

- Vehicle concentration headline: Top 10 vehicles hold 27.28% of exposure (Top 1: 7.48%).

Operational actions

- Cap exposure per vehicle (per-day / per-trip).
- Require **secondary vehicle readiness** for routes where Top-1 share is extreme.
- Telemetry + photo-verification triggers for high-exposure vehicles.

5. Carrier Risk Profile

Top 10 carriers by exposure

carrier_norm	trips	total_exposure	avg_exposure	unique_routes
:---:	---:	---:	---:	---:
SHYAMSUNDER BHIMSAIN GOYAL	68	6.00887e+08	8.83657e+06	18
YATAYAT CORPORATION OF INDIA	20	1.45967e+08	7.29833e+06	14
INLAND WORLD LOGISTICS PRIVATE LIMITED	11	9.90046e+07	9.00042e+06	7
RAVINDER SINGH	8	7.07825e+07	8.84782e+06	5
PREMIER ROAD CARRIERS LIMITED	3	3.22228e+07	1.0741e+07	1
SURAT GOODS TRANSPORT PVT LTD	2	2.62845e+07	1.31422e+07	1
NAVEEN S DADHICH	6	2.18987e+07	3.64979e+06	4
EITA INDIA LIMITED	2	2.12872e+07	1.06436e+07	1
ADHUNIK TRANSPORT ORGANISATION LIMITED	2	1.22012e+07	6.1006e+06	1
Surat Goods transport	1	8.7615e+06	8.7615e+06	1

Carrier × Route hotspots

Use the carrier-route concentration summary to identify where underwriting terms should be carrier-specific.

carrier_norm	route	pair_trips	pair_total_exposure	pair_unique_vehicles
:----------	:----------	----------:	----------:	----------:
SHYAMSUNDER BHIMSAIN GOYAL	Vadodara → Nalgonda	13	1.56975e+08	11
SHYAMSUNDER BHIMSAIN GOYAL	Vadodara → Surat	16	1.1522e+08	14
SHYAMSUNDER BHIMSAIN GOYAL	Vadodara → Visakhapatnam	5	6.78568e+07	4
SHYAMSUNDER BHIMSAIN GOYAL	Vadodara → Bangalore	6	5.21963e+07	6
SHYAMSUNDER BHIMSAIN GOYAL	Vadodara → East Godavari	4	4.1106e+07	3
INLAND WORLD LOGISTICS PRIVATE LIMITED	Vadodara → Raipur	2	3.58793e+07	2
YATAYAT CORPORATION OF INDIA	Vadodara → South West Delhi	2	3.34233e+07	2
PREMIER ROAD CARRIERS LIMITED	Vadodara → Bangalore	3	3.22228e+07	3
SHYAMSUNDER BHIMSAIN GOYAL	Vadodara → Cuttack	2	3.1152e+07	2
RAVINDER SINGH	Vadodara → Raigarh(MH)	2	3.0529e+07	2

What to demand from each carrier tier

- **Tier RED (highest exposure / hotspots):** route compliance evidence, vehicle fitness proof, stricter SLAs, higher deductibles.
- **Tier YELLOW:** periodic route audit, standard telemetry/photo checks.
- **Tier GREEN:** keep as preferred list, reward with smoother terms.

6. Consignee Risk Profile

Top 10 consignees by exposure

consignee_norm	trips	total_exposure	unique_carriers
:----------	--------:	----------:	----------:
DIVI S LABORATORIES LIMITED	10	1.2655e+08	1
KARTIK REFRIGERATION PRIVATE LIMITED	11	1.22376e+08	2
VOLTAS LIMITED	14	9.54345e+07	4
BANGALORE METRO RAIL CORPORATION LIMITED	9	8.44192e+07	2
DIVIS LABORATORIES LIMITED	6	6.5034e+07	1
KMV PROJECTS LIMITED	4	5.62034e+07	3
SRF LIMITED	3	3.5046e+07	2
DELHI INTERNATIONAL AIRPORT LIMITED	2	3.34233e+07	1
BLUE JET HEALTHCARE LIMITED	2	3.0529e+07	1
CEMINDIA PROJECTS LIMITED	2	2.96982e+07	1

Consignee × Carrier clusters

Use the hotspot plot/CSV to prioritize joint interventions (who ships with whom at scale).

- CSV: `L2_consignee_carrier_hotspots__20260402_191659.csv`
- Figures (if present): `L2_consignee_x_carrier_hotspot__20260402_191659.png`

Why consignees should co-fund risk reduction

Consignees are the stable node in the system. Carriers and drivers churn; consignees don't. Funding telemetry, secure yards, unloading discipline, and route compliance is cheaper than absorbing disruption.

7. What This Enables

- **Pricing leverage:** rate adjustments based on exposure concentration (not gut feel).
- **Deductible rationalisation:** higher deductibles where operational stress concentrates.
- **Route exclusions:** defendable route carve-outs when exposure/risk spine is extreme.
- **Embedded insurance logic:** integrate operational proofs (route/vehicle controls) with underwriting terms.

8. What This Does NOT Claim

- This is **not** a claims predictor.
- This is **not** a loss probability model (yet).
- That honesty makes this defensible: it's a **risk concentration + operational stress intelligence layer** that underwriters can audit and act on.

Appendix — Key artifacts generated

- Route scorecard CSV: `L1_route_scorecard__20260402_191659.csv`
- Route spine CSV: `L1_route_spine__20260402_191659.csv`
- Vehicle concentration CSV: `L1_vehicle_concentration__20260402_191659.csv`
- Carrier exposure CSV: `L2_carrier_exposure__20260402_191659.csv`
- Consignee exposure CSV: `L2_consignee_exposure__20260402_191659.csv`

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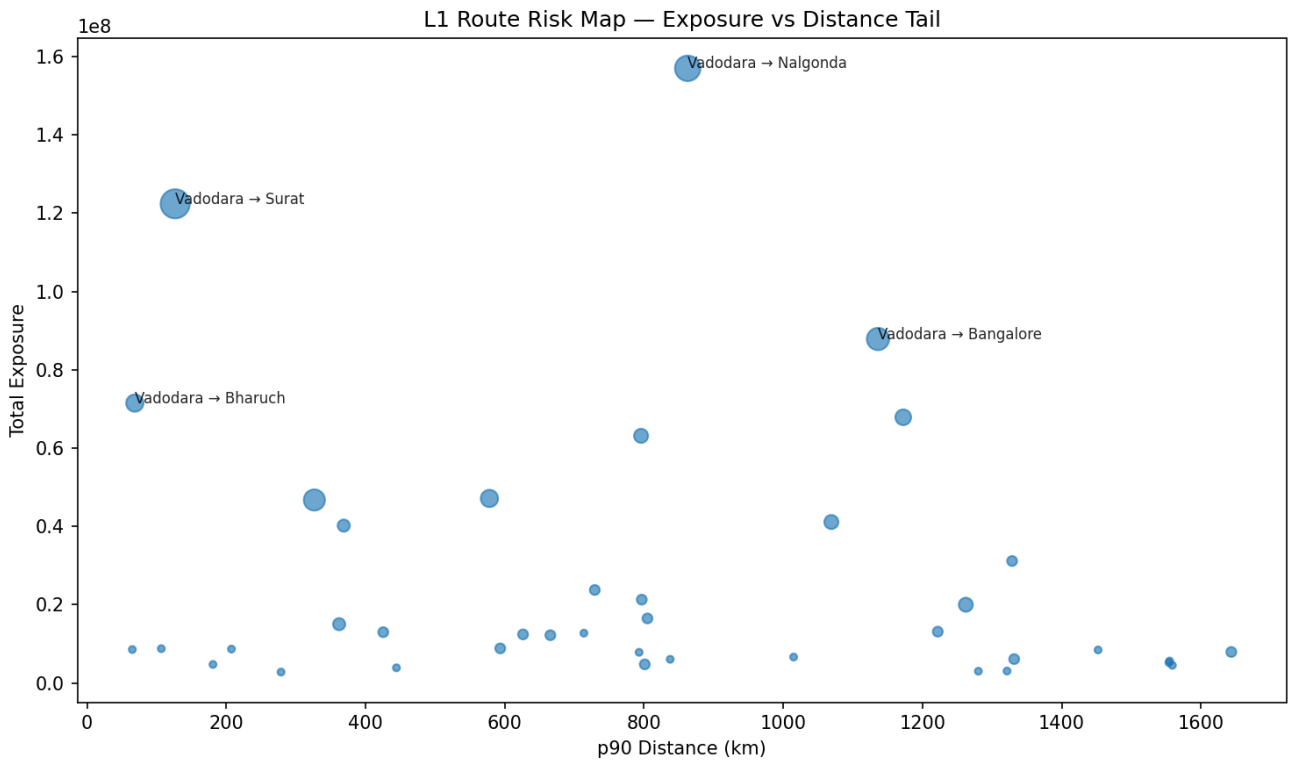
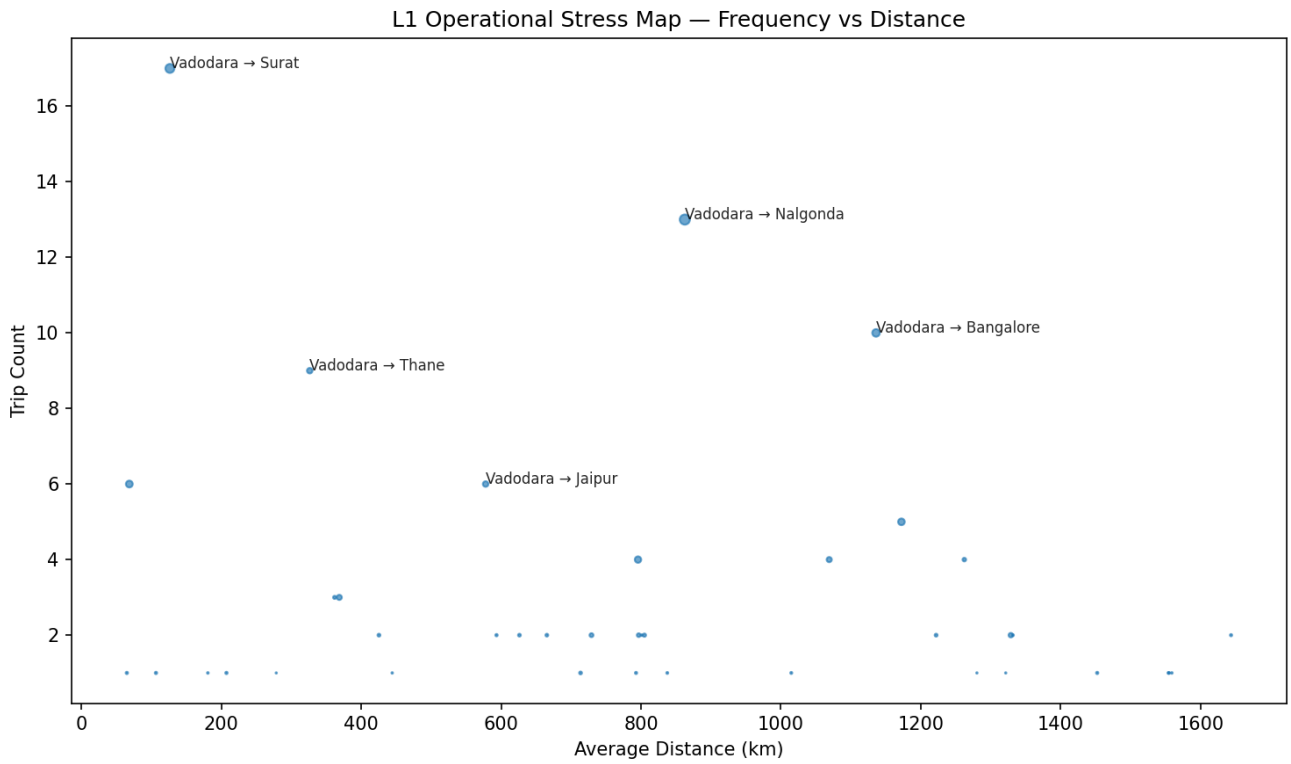
Visual Analytics Plates

L1 route risk map



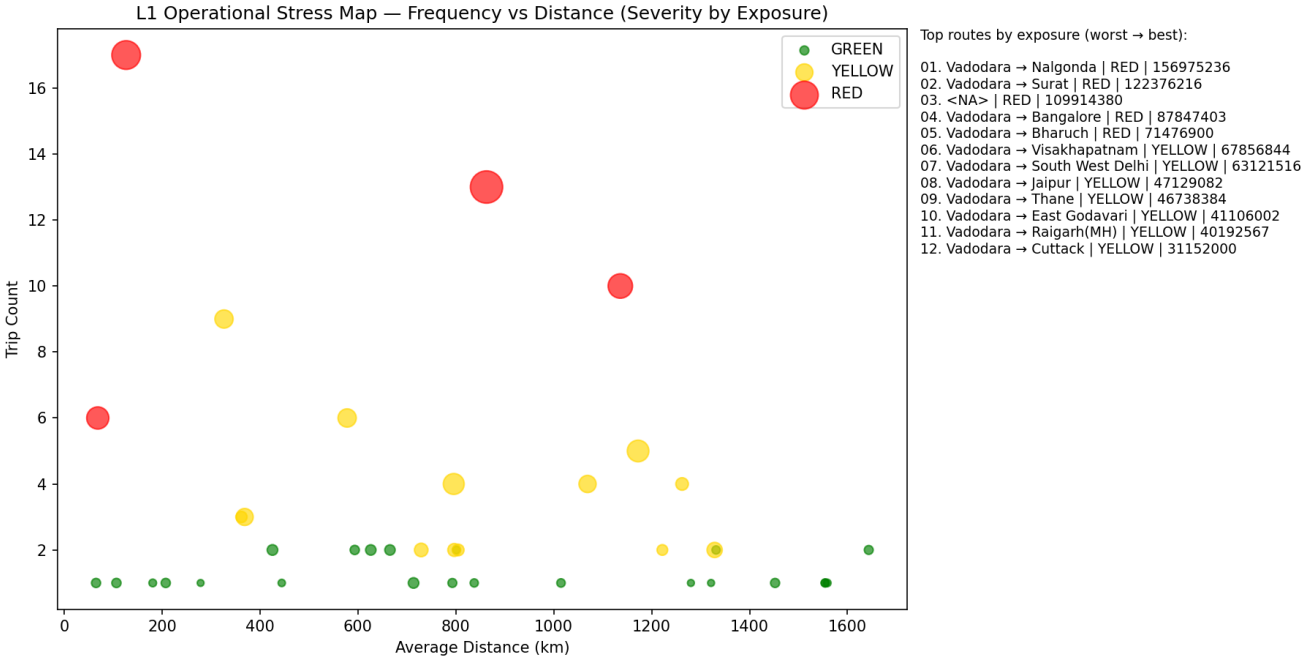
L1 route risk map.png

L1 operational stress map



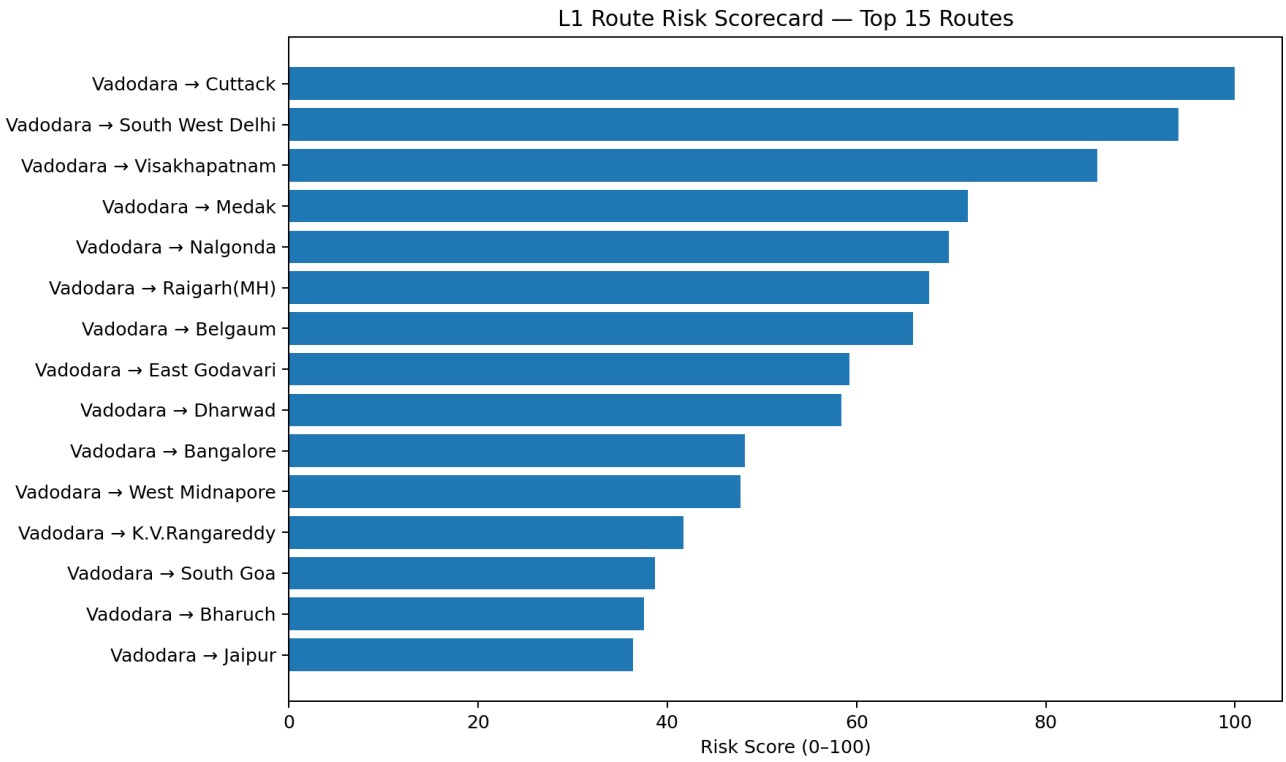
L1 operational stress map.png

L1 operational stress map sidepanel



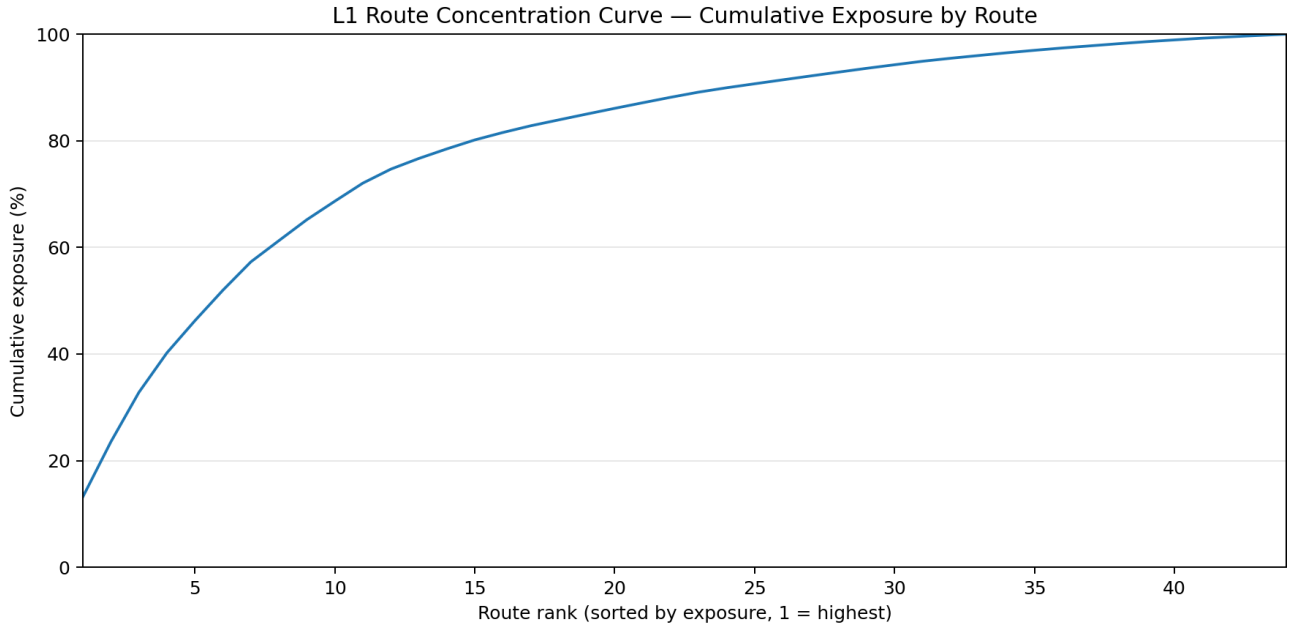
L1 operational stress map sidepanel.png

L1 route scorecard top15



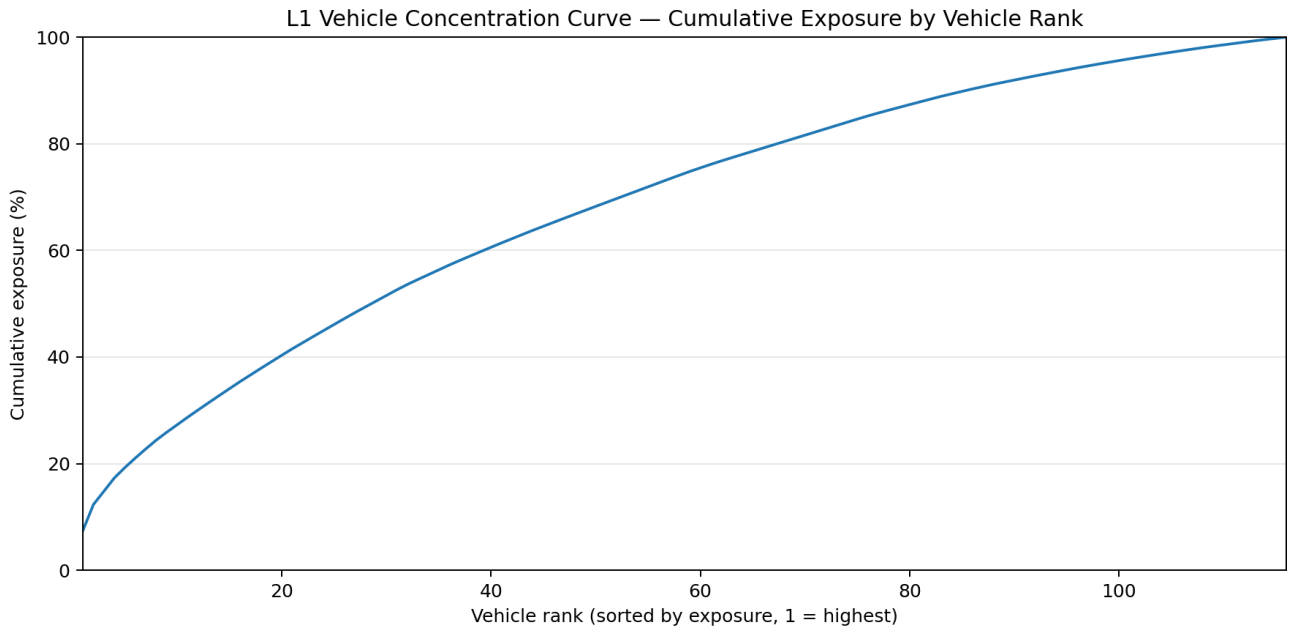
L1 route scorecard top15.png

L1 route concentration curve



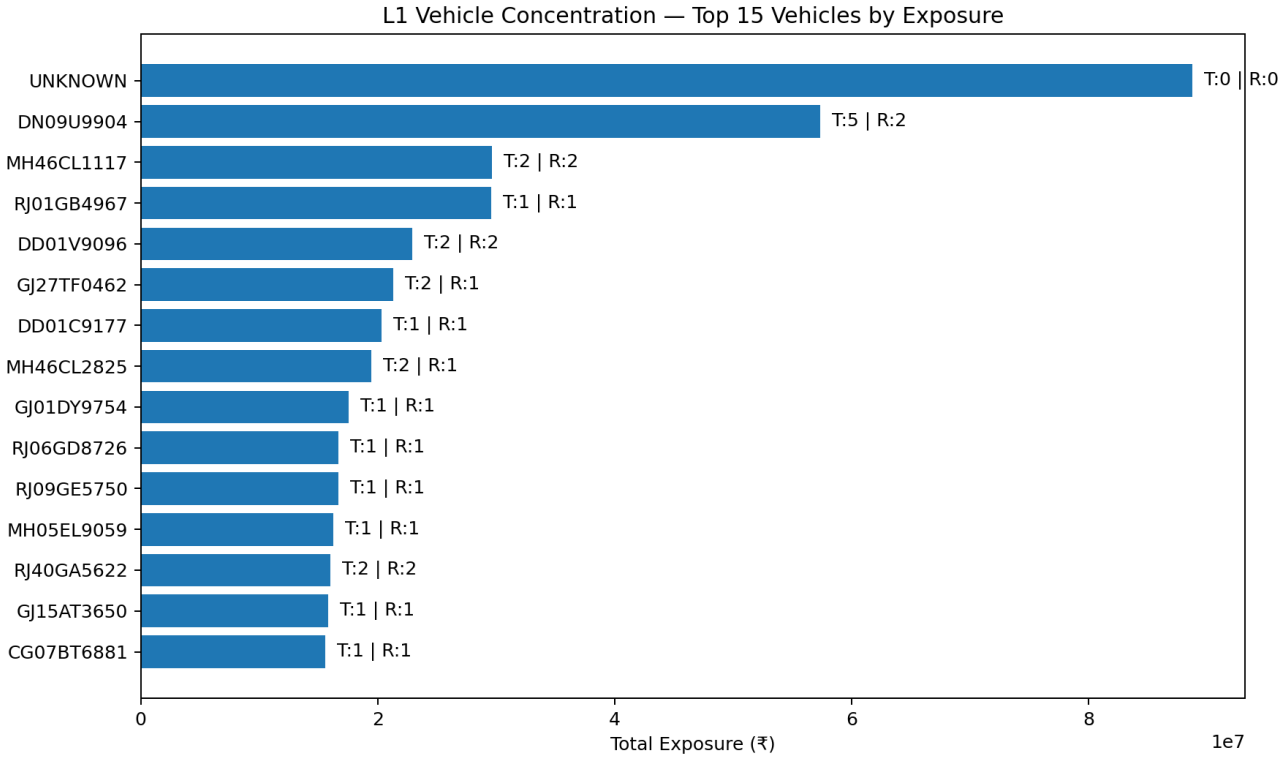
L1 route concentration curve.png

L1 vehicle concentration curve



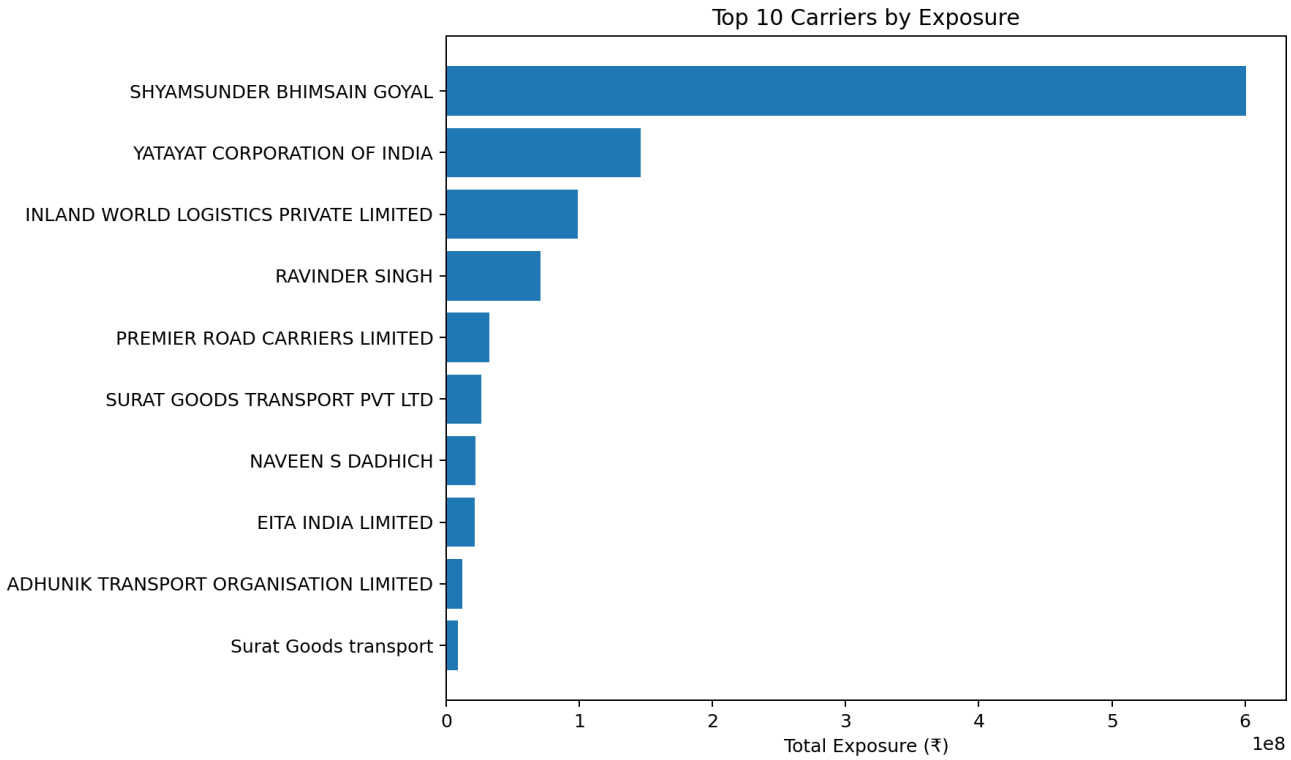
L1 vehicle concentration curve.png

L1 vehicle concentration top15



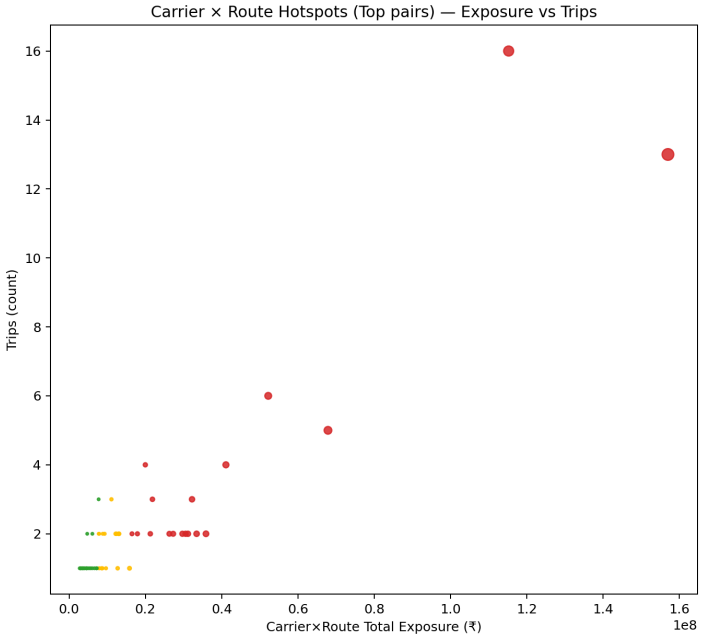
L1 vehicle concentration top15.png

L2 carrier top10



L2 carrier top10.png

L2 carrier x route hotspot

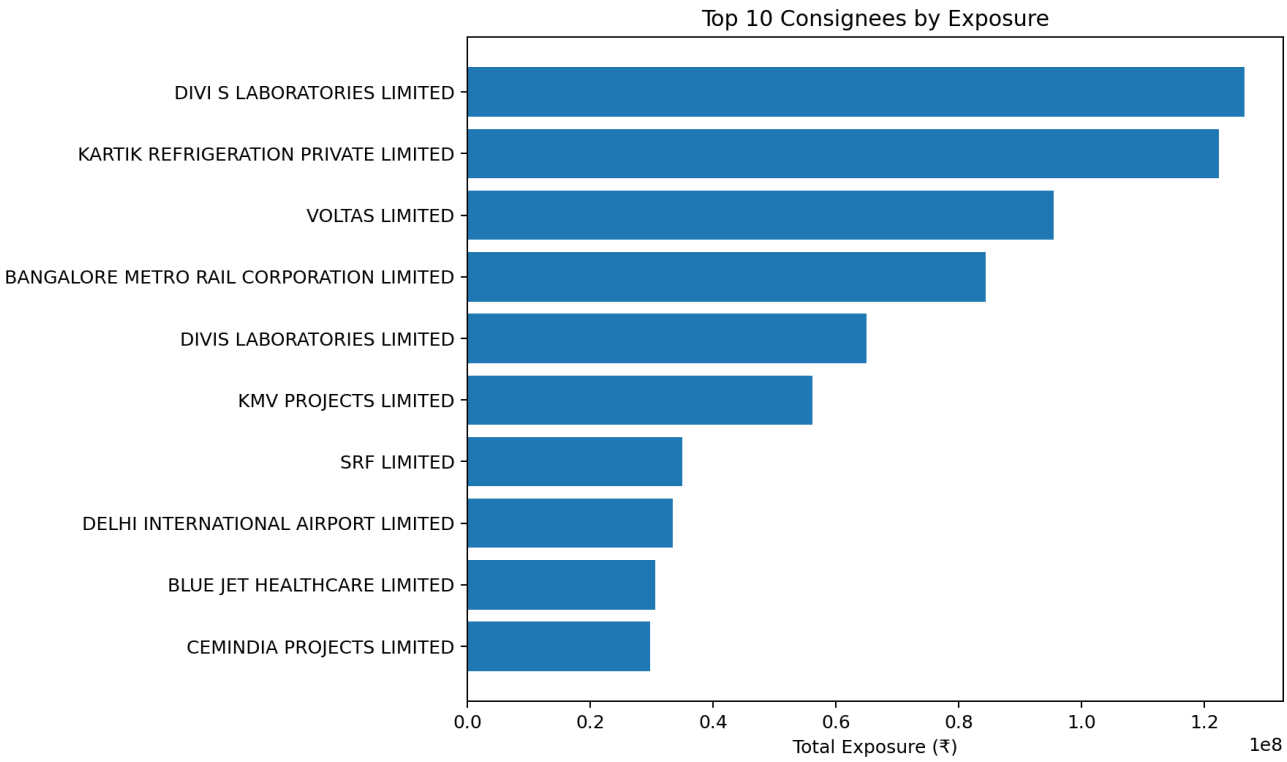


Top pairs (ranked by exposure):

01. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Nalgonda | RED | ₹156,975,236 | T:13
02. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Surat | RED | ₹115,220,472 | T:16
03. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Visakhapatnam | RED | ₹67,856,844 | T:5
04. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Bangalore | RED | ₹52,196,346 | T:6
05. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → East Godavari | RED | ₹41,106,002 | T:4
06. INLAND WORLD LOGISTICS PRIVATE LIMITED | Vadodara → Jaipur | RED | ₹35,879,316 | T:2
07. YATAYAT CORPORATION OF INDIA | Vadodara → South West Delhi | RED | ₹33,423,276 | T:2
08. PREMIER ROAD CARRIERS LIMITED | Vadodara → Bangalore | RED | ₹32,222,850 | T:3
09. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Cuttack | RED | ₹31,152,000 | T:2
10. RAVINDER SINGH | Vadodara → Raigarh(MH) | RED | ₹30,528,957 | T:2
11. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → South West Delhi | RED | ₹29,698,240 | T:2
12. INLAND WORLD LOGISTICS PRIVATE LIMITED | Vadodara → Bharuch | RED | ₹27,280,000 | T:2
13. SURAT GOODS TRANSPORT PVT LTD | Vadodara → Bharuch | RED | ₹26,284,500 | T:2
14. YATAYAT CORPORATION OF INDIA | Vadodara → Thane | RED | ₹21,842,744 | T:3
15. EITA INDIA LIMITED | Vadodara → Dharwad | RED | ₹21,287,199 | T:2
16. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Kanchipuram | RED | ₹19,982,710 | T:4
17. YATAYAT CORPORATION OF INDIA | Vadodara → Bharuch | RED | ₹17,912,400 | T:2
18. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → K.V.Rangareddy | RED | ₹16,484,600 | T:2
19. RAVINDER SINGH | Vadodara → Belgaum | YELLOW | ₹15,835,600 | T:1
20. INLAND WORLD LOGISTICS PRIVATE LIMITED | Vadodara → Angul | YELLOW | ₹13,109,800 | T:2
21. YATAYAT CORPORATION OF INDIA | Vadodara → Pune | YELLOW | ₹12,970,779 | T:2
22. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Medak | YELLOW | ₹12,726,890 | T:1
23. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Ananthapur | YELLOW | ₹12,408,880 | T:2
24. ADHUNIK TRANSPORT ORGANISATION LIMITED | Vadodara → Alwar | YELLOW | ₹12,201,200 | T:2
25. RAVINDER SINGH | Vadodara → Thane | YELLOW | ₹11,097,900 | T:3
26. YATAYAT CORPORATION OF INDIA | Vadodara → Raigarh(MH) | YELLOW | ₹9,663,610 | T:1
27. NAVEEN S DADHICH | Vadodara → Thane | YELLOW | ₹9,319,640 | T:2
28. SHYAMSUNDER BHIMSAIN GOYAL | Vadodara → Sagar | YELLOW | ₹8,833,150 | T:2
29. Surat Goods transport | WAGHODIA → DAHEJ | YELLOW | ₹8,761,500 | T:1
30. CHAUHAN PRADEEPSINH | Vadodara → Valsad | YELLOW | ₹8,664,150 | T:1
31. YATAYAT CORPORATION OF INDIA | Vadodara → Kheda | YELLOW | ₹8,539,879 | T:1

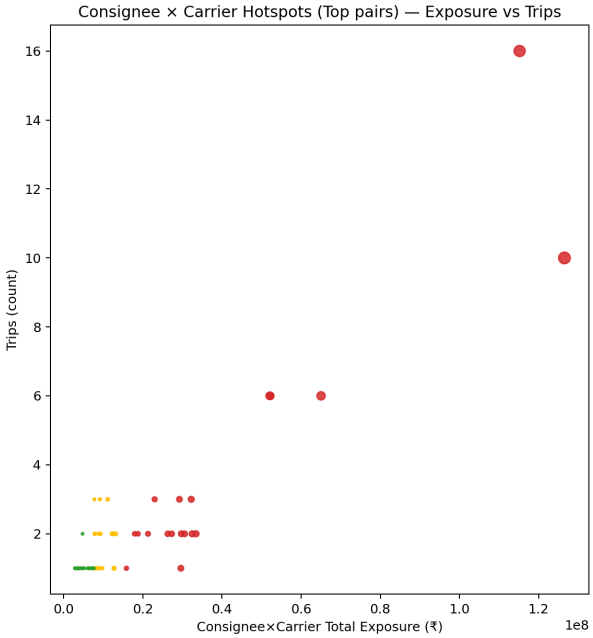
L2 carrier x route hotspot.png

L2 consignee top10



L2 consignee top10.png

L2 consignee x carrier hotspot

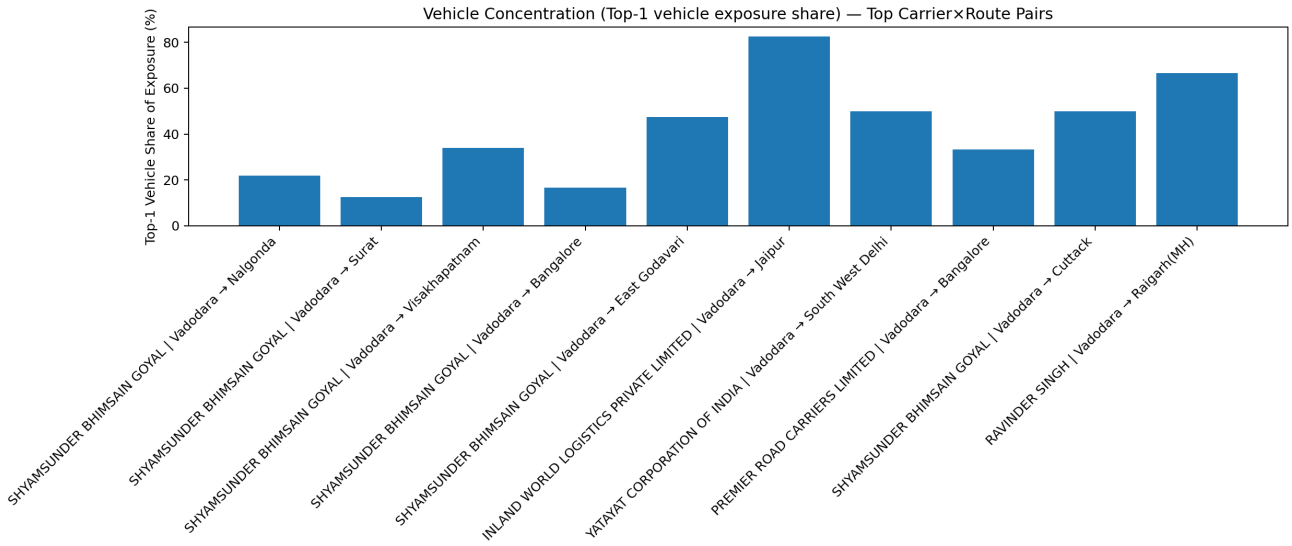


Top pairs (ranked by exposure):

01. DIVI S LABORATORIES LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | RED | ₹126,549,808 | T:10
02. KARTIK REFRIGERATION PRIVATE LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | RED | ₹115,220,472 | T:16
03. DIVIS LABORATORIES LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | RED | ₹65,034,002 | T:6
04. BANGALORE METRO RAIL CORPORATION LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | RED | ₹52,196,346 | T:6
05. VOLTAS LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | RED | ₹52,043,310 | T:6
06. DELHI INTERNATIONAL AIRPORT LIMITED | YATAYAT CORPORATION OF INDIA | RED | ₹33,423,276 | T:2
07. KMV PROJECTS LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | RED | ₹32,450,000 | T:2
08. BANGALORE METRO RAIL CORPORATION LIMITED | PREMIER ROAD CARRIERS LIMITED | RED | ₹32,222,850 | T:3
09. BLUE JET HEALTHCARE LIMITED | RAVINDER SINGH | RED | ₹30,528,957 | T:2
10. CEMINDIA PROJECTS LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | RED | ₹29,698,240 | T:2
11. SHRI KALYAN BUILDMART PRIVATE LIMITED | INLAND WORLD LOGISTICS PRIVATE LIMITED | RED | ₹29,594,400 | T:1
12. RENAATUS PROJECTS PRIVATE LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | RED | ₹29,231,898 | T:3
13. AARTI INDUSTRIES LIMITED | INLAND WORLD LOGISTICS PRIVATE LIMITED | RED | ₹27,280,000 | T:2
14. SRF LIMITED | SURAT GOODS TRANSPORT PVT LTD | RED | ₹26,284,500 | T:2
15. VOLTAS LIMITED | YATAYAT CORPORATION OF INDIA | RED | ₹22,973,639 | T:3
16. AIRPORTS AUTHORITY OF INDIA | EITA INDIA LIMITED | RED | ₹21,287,199 | T:2
17. LUPIN LIMITED | YATAYAT CORPORATION OF INDIA | RED | ₹18,756,100 | T:2
18. GREEN LEAF CHEM TECH PRIVATE LIMITED | YATAYAT CORPORATION OF INDIA | RED | ₹17,912,400 | T:2
19. KMV PROJECTS LIMITED | RAVINDER SINGH | RED | ₹15,835,600 | T:1
20. ADVANCE VENTILATION PRIVATE LIMITED | INLAND WORLD LOGISTICS PRIVATE LIMITED | YELLOW | ₹13,109,800 | T:2
21. NEULAND LABORATORIES LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | YELLOW | ₹12,726,890 | T:1
22. DIVIS LABORATORIES LTD | SHYAMSUNDER BHIMSAIN GOYAL | YELLOW | ₹12,672,374 | T:1
23. GROB MACHINE TOOLS INDIA PRIVATE LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | YELLOW | ₹12,408,880 | T:2
24. MORIROKU TECHNOLOGY INDIA PRIVATE LIMITED | ADHUNIK TRANSPORT ORGANISATION LIMITED | YELLOW | ₹12,201,200
25. VOLTAS LIMITED | RAVINDER SINGH | YELLOW | ₹11,097,900 | T:3
26. PRIVI SPECIALITY CHEMICALS LIMITED | YATAYAT CORPORATION OF INDIA | YELLOW | ₹9,663,610 | T:1
27. VOLTAS LIMITED | NAVEEN S DADHICH | YELLOW | ₹9,319,640 | T:2
28. SARAVANA STORES CELEBRITY PRIVATE LIMITED | NAVEEN S DADHICH | YELLOW | ₹9,150,900 | T:3
29. TRENT HYPERMARKET PRIVATE LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | YELLOW | ₹8,923,750 | T:1
30. BHARAT PETROLEUM CORPORATION LIMITED | SHYAMSUNDER BHIMSAIN GOYAL | YELLOW | ₹8,833,150 | T:2
31. SRF LIMITED | Surat Goods transport | YELLOW | ₹8,761,500 | T:1

L2 consignee x carrier hotspot.png

L2 vehicle concentration top1



L2 vehicle concentration top1.png